

Overview of Valuation

Chapter Overview

This chapter focuses on the inherent challenges that arise in the valuation of investments, from large capital budgeting projects to the acquisition of stand-alone businesses. First, we review the notion of project valuation in terms of its anticipated impact on the wealth of the firm's owners. Second, using a case study involving a large investment made by a group of multinational oil companies, we highlight five key challenges that can arise in valuing a major investment proposal.

To deal effectively with the challenges involved in valuing major investments, firms must have a disciplined approach, founded on a sound evaluation process. We present a three-phase investment evaluation process to address this need. This process begins with the identification of an investment idea and ends with the final approval. This process does not eliminate all bad investments, for investing is inherently risky. However, it does help ensure that the firm does not fall victim to decision errors based on flawed analysis.

1 INTRODUCTION

Valuation is central to most of what we do in financial analysis. When firms evaluate either an internally generated investment project or an external acquisition, one of the first steps in the process is a valuation of the opportunity. Moreover, when firms consider either share issuances or share repurchases, an initial step in the evaluation process is the valuation of the firm's own shares.

To illustrate the key role played by valuation in financial decisions, let's first consider Starbucks' (SBUX) acquisition of Seattle's Best Coffee in 2006. As part of the acquisition process, Starbucks' financial analysts estimated the intrinsic value of the shares of Seattle's Best in order to determine how much they would be willing to pay for the company. Starbucks' management probably also did a valuation of their own shares to gauge whether their stock was under- or overvalued, because this would influence whether they would want to make an all-cash offer for Seattle's Best or offer Starbucks shares in exchange for Seattle's Best shares.

If you were a security analyst responsible for advising investors on whether or not to buy or sell Starbucks common stock, you would also be keenly interested in this

acquisition. Your job would require you not only to value each of the companies, but also to assess whether their combination would generate new sources of value (i.e., synergies) or be an enormous waste of resources.

Corporate financial analysts also spend considerable amounts of time in valuing investment projects. For example, before introducing its new Via[®] product line of instant coffees, Starbucks' financial analysts performed a valuation of the new product line that included an analysis of the demand for the product, as well as the costs of developing, marketing, and producing the new product line. The final go–no go decision on Via[®] boiled down to a comparison of the value of the investment's future cash flows with the cost of launching the investment.

Whether you are valuing an ongoing business or an investment project, the procedure you follow is essentially the same. In each case, you will need lots of information. Specifically, you will need to

1. make estimates of the cash flows generated by the investment.
2. assess the riskiness of those cash flows and determine the appropriate discount rate.
3. identify comparable investments that are either publicly traded or have recently been bought or sold.

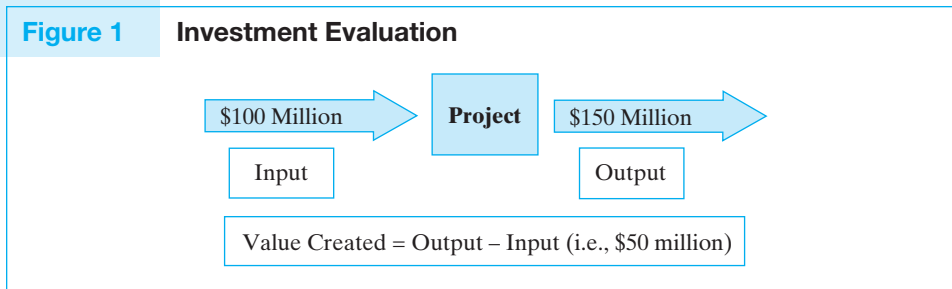
If you are valuing the shares of a publicly traded firm, the process is relatively straightforward. The company will have a history that includes past cash flows and stock returns, and in most cases, you will have access to information about other publicly traded firms that are comparable. The information about comparable firms will prove to be very helpful in estimating the value of the firm of interest. The valuation of an investment project can be more challenging because you do not typically have the benefit of historical data and are likely to have fewer comparable transactions that you can use to help gauge how other market participants have evaluated similar investments. However, the valuation process in each case is very much the same.

As a financial analyst evaluating a potential acquisition, you are faced with additional challenges. In addition to assessing the cash flows of the acquisition candidate as a stand-alone company, you will want to identify any synergies with the acquiring firm that make the value of the combined firms greater than the sum of the parts.

2 THE NATURE OF MAJOR INVESTMENT DECISIONS

As the above discussion indicates, firms grow and expand their operations in one of two ways: They acquire productive capacity by assembling the necessary assets, or, alternatively, they acquire an existing business. In the first case, we refer to the valuation problem as **project valuation**, and in the second case, we refer to the valuation problem as **enterprise valuation**. Throughout this book, we will consider examples of both project and enterprise valuation and provide a common set of tools and underlying principles that can be used for both types of analysis.

Throughout most of the book, we will assume that the firm's objective is to create wealth by initiating and managing investments that generate future cash flows that are worth more (i.e., have a higher monetary value) than the costs of the investment. On the surface, the process appears quite simple. Consider the illustration in Figure 1:

Figure 1 Investment Evaluation

A firm has the opportunity to invest \$100 million in a project that generates a stream of cash flows that are valued at \$150 million. By making the investment, the firm provides an incremental \$50 million in wealth for its shareholders. In the language of the financial analyst, the firm expects the project to be worth \$50 million more than it costs (in today's dollars). Practically speaking, the firm expects to make its shareholders \$50 million. In the jargon of finance, the project has an expected **net present value** (NPV) of \$50 million.

It is, however, an unfortunate fact of life that over half of all large investment projects fail to achieve their hoped-for results.¹ The record of merger and acquisition performance is even more problematic, as exemplified by Daimler-Benz's purchase of Chrysler for \$35 billion in 1998. Chrysler was later sold in 2007 for only \$7.4 billion and then went bankrupt and was taken over by Fiat in 2009. If project evaluation is as straightforward as depicted in Figure 1, then why do so many big investments fail?

One possible explanation for failed investments is simply that firms invest in risky projects, and we should not expect them to be right all the time. Using a sports analogy, when you swing for the fences, you can expect to strike out a lot. However, we will contend throughout this book that there is more than investment risk and bad luck at work here. The fact is that analyzing capital expenditure choices can be both complex and tedious, and managers must make their investment decisions on the basis of incomplete information about uncertain future events. In the face of this complexity and uncertainty, managers often just "go with their gut" and initiate the investments that feel right.

We agree that managers with reliable intuition (that is, a discriminating gut) and the experience to make sound judgments will—and should—ultimately make the major corporate investment choices. However, analytical tools, as well as sophisticated yet inexpensive and user-friendly computer software, can help managers see through the complexity and work through the tedium inherent in the evaluation of a major investment. It is our belief that by using these tools and taking a more disciplined approach to valuation, managers' judgment will be enhanced and they will make better investment choices.

¹ Matta and Ashkenas (2003) arrive at this conclusion by analyzing a set of project initiatives that include major technology installations, post-merger integrations, and new-growth strategies (Nadim F. Matta and Ronald N. Ashkenas, "Why Good Projects Fail Anyway," *Harvard Business Review* [September 2003]: 109–14).

3 VALUING PROJECTS AND BUSINESSES

To provide a brief overview of the kinds of issues that we will be exploring in this book, this section will describe two major investments. The first is a project that resulted in the development of the Caspian Sea oil fields, which are currently a major source of oil. The second is the acquisition of Linksys Group by Cisco Systems (CSCO).

Project Valuation—Investing in the Caspian Sea Oil Fields

To illustrate the challenges associated with valuing a large investment opportunity, we will start by describing a major oil investment in the Caspian Sea.² Although this investment can be thought of as an investment project, it was in fact the creation of a new firm that was jointly held by lots of oil companies in the early 1990s. Take yourself back in time, and imagine that you are sitting at the table when the analysis begins and are trying to get a handle on the risks and potential rewards that might accrue to your firm if it undertakes the investment.

To be more specific, the investment consists of part ownership of a joint-venture consortium comprised of 11 oil companies. The consortium, known as the Azerbaijani International Oil Consortium, includes the state oil company of Azerbaijan, British Petroleum, Amoco, the national oil companies of Russia and Turkey, and several other foreign oil companies. The 11 oil companies jointly have the right to develop three oil fields in the western part of the Caspian Sea that were estimated to contain 4.5 to 5 billion barrels of oil.³

The right to develop the three fields would be subject to certain conditions. The consortium would complete a seismic survey, an environmental impact study, and the drilling of a series of test wells. It would then have to submit a development plan to the state oil company of Azerbaijan, outlining the development of the fields based on its preliminary findings. Following the initial work, the consortium would eventually submit a plan involving four stages, summarized in Figure 2. Both the state oil company of Azerbaijan and the consortium would have the right to approve each step in the process based on the results of the prior stage. Furthermore, a production-sharing agreement would define the revenue-sharing agreement for the output from the investment, if it were successful.

As the results of each stage of the project would become known, the management of the state oil company of Azerbaijan and the consortium would be faced with the decision of whether they should enter the next phase of the project. In essence, if the decision were made to proceed with the early-stage investment, the consortium would acquire the *option* to make successive investments in each of the three subsequent stages of the development process. Furthermore, assuming that all stages of the investment were successfully executed, the fact that the consortium had developed expertise in the region would make it possible for each member of the consortium to compete on favorable terms with would-be competitors for future investment opportunities.

² For more information, see Benjamin Esty and Michael Kane, BP Amoco (B): Financing Development of the Caspian Oil Fields, *Harvard Business School Press*, Case #9-201-067, 2001.

³ “Azerbaijan-Pipeline Knocked Back,” *Project Finance International* (March 24, 1999): 45.

Figure 2 Stages of Investment in the Caspian Sea Oil Development Investment

1998–1999	Early Oil Project
	■ Develop Chirag field by refurbishing offshore drilling platform and drilling new wells. ■ Construct undersea oil pipeline to terminal (105 miles). ■ Rebuild two export pipelines (total of 1,300 miles). ■ Estimated cost = \$1 billion.
Beginning in 2000	First-Stage Development
	■ Develop the Azeri field. ■ Estimated cost = between \$2.6 and \$3.1 billion.
Beginning in 2002	Second-Stage Development
	■ Develop the deepwater Gunashli field. ■ Estimated cost = \$3 billion.
Beginning in 2003–2004	Third-Stage Development
	■ Additional development of the Azeri field. ■ Estimated cost = \$2 billion.

Issues to Consider When Valuing an Investment

The Caspian Sea oil-field development project illustrates the complexities of the environment in which companies make investment decisions. In any situation in which a company must value a major new investment, five key issues arise. Figure 3 highlights these issues, which we discuss below.

Issue #1: Does the “Story” Make Sense?

Before the firm makes any investment, the investment story, or strategy, must be plausible to decision makers. By “make sense,” we mean that management must be convinced that the potential gains from the investment are large enough to warrant initial investigation. Also,

Figure 3 Important Issues to Think About When Making a Major Investment

1. Does the “story” make sense?
2. What are the risks involved in undertaking the investment?
3. How can the investment be financed?
4. How does the investment affect near-term earnings?
5. Does the investment have inherent flexibilities?

management must believe that the firm's management team possesses (or can acquire) the expertise required to reap the rewards of investing. In the terminology of economics, the issue is *competitive advantage*. Specifically, does the firm have an advantage due to specialized knowledge or circumstance that allows it to capture the benefits of the investment?

The ultimate success or failure of an investment is largely driven by the capabilities of the firm that undertakes the investment in comparison to the capabilities of other firms. This notion is captured in the concept of *comparative advantage*. In addition, the value that the firm is able to capture for itself is affected by the actions and reactions of competitor firms. Therefore, a complete analysis of an investment must address the following strategic issues as part of its "story": Does the firm, and does this project, provide any comparative advantage(s) relative to those of other firms and existing projects? Are the firm's comparative advantages sufficient to deter competitors from making similar investments?

Issue #2: What Are the Risks Entailed in the Investment, and How Can They Be Assessed and Dealt with in the Analysis?

The old adage "Look before you leap" is good advice when evaluating investment opportunities. Specifically, a careful assessment of *what might go wrong* is perhaps even more important than an analysis of what we hope will go right. For example, international investment projects that are made in emerging markets expose the investing company to a myriad of risks.

Here are some questions that often come up in evaluating and managing risk: What are the underlying risks associated with the investment? How should these risks be incorporated into the project analysis? Do the investment risks affect the rate of return that should be used in evaluating whether to undertake the investment? Are there governmental programs (domestic or foreign) that can insure the investment in the event of political instability? How does the ability of the firm to transfer investment risk affect the financing of the project and the project's valuation?

Issue #3: How Can the Investment Be Financed?

There are a variety of ways to fund a major investment project. Moreover, the ability to secure attractive financing is a key determinant of the value of the investment. For example, sometimes firms can obtain debt subsidies in the form of governmental guarantees or credit enhancements, especially when international investments are involved. In other cases, firms might raise private equity and new debt financing to fund off-balance-sheet investments known as special-purpose entities (SPEs).

Specifically, the evaluation of financing opportunities addresses issues such as the following: How do the characteristics of the firm and the project (e.g., the extent that the risks of the project can be managed by transferring the risk to another party via a financial contract) affect how it is financed? How does the financing of the project affect how the project can be valued?

In addition to asking how much debt to use, the firm should also discuss the type of debt to use. Should the project be financed on the firm's balance sheet, or should it be financed off the balance sheet with nonrecourse debt (i.e., project financing)?⁴

⁴ *Nonrecourse* or project financing refers to debt for which only the investment or project is responsible. That is, the debt is to be repaid out of the cash flows generated by the project, and there is no recourse to the company sponsoring the investment.

Issue #4: How Does the Investment Affect Near-Term Earnings?

Investors and equity analysts use the firm's reported earnings as an indicator of the firm's success or failure. When considering a large investment, managers will be keenly aware of its effect on earnings. They will ask whether a project is likely to *dilute* (reduce) or *accrete* (increase) the firm's earnings per share. For example, a major oil project like the Caspian Sea project can often dilute the firm's earnings in the early years because of the considerable up-front expenses and deferred cash flows. However, earnings should increase over time as the future benefits from the project materialize.

A project's effect on earnings can be important for a variety of reasons in determining whether managers are willing to initiate an investment. For example, if company executives are being paid on earnings-based performance measures, or if they believe that investors focus on earnings per share, then they may be reluctant to invest in a project that negatively affects the firm's earnings. Thus, accounting and the design of pay-for-performance policies often influence a firm's investment decisions. It is important to investigate the potential importance of earnings on a firm's investment decisions, as well as how residual income (or Economic Value AddedTM)⁵ can be used to help resolve the problems associated with earnings as a performance metric.

Issue #5: Does the Investment Have Inherent Flexibilities That Allow the Firm to Modify It in Response to Changing Circumstances?

Firms make investments expecting that a particular set of outcomes will guide the investment toward a particular result. However, uncertain future events make it particularly important that the project provide opportunities to react and adapt the investment to changing circumstances. Specifically, these opportunities include the following.

Can the investment be staged? Often, as in the Caspian Sea oil-fields case, firms undertake very large investments in stages. Staging allows the firm to manage its risk exposure by making a series of successively larger commitments based upon the success of the prior investment. This is clearly the case with large oil-development projects: They typically have an early stage that provides information about the size of the oil reserves before the firm initiates later development stages. This is also true for many new products or services for which the test-marketing phase reveals important information about their sales potential.

When the firm invests in the initial stages, it essentially acquires the "option" to invest in later stages of the project (if the intermediate investments prove fruitful). The flexibility to delay implementation of a project—to cut one's losses and abandon a project or to expand a successful investment—are examples of optionality that can add considerable value to a project.

Decision makers should be aware that, in some instances, what appears to be an option turns out to be an obligation. Specifically, if there is no real possibility that an option will *not* be exercised, then it's not an option at all. In the Caspian Sea case, for example, it may turn out to be very difficult for the consortium members to back out after stage 1.

⁵ Economic Value Added or EVATM is a registered trademark of Stern Stewart and Company.

Does the investment offer the opportunity for follow-on investments? The opportunity to invest in a new product, market, or technology can provide valuable “follow-on investment” opportunities. Most new investment opportunities arise out of previous investments that the firm has made. In the Caspian Sea case, there are likely to be additional oil and gas deals, or even the opportunity to enter the petrochemical business in central Asia. Consequently, the valuation of investments with follow-on opportunities requires consideration of *two sets* of cash flows: the cash flows provided by the immediate opportunity as well as those from the possible subsequent projects. The fundamental issue is this: How should the company value these follow-on opportunities and incorporate them into the analysis of the initial decision to invest?

Does the investment provide production or marketing synergies with existing products? To the extent that the new investment shares existing production and/or marketing resources, the opportunity exists to gain a comparative advantage over the firm’s competitors.

Enterprise Valuation—Mergers and Acquisitions

The five basic issues encountered in analyzing the Caspian Sea investment project also apply to *enterprise valuation*—the valuation of an entire firm. From the financial analyst’s perspective, the fundamental question is the same: What is the investment worth, and how does this compare to its cost?

Cisco Systems (CSCO) is noted for its policy of acquiring existing firms. This strategy of expansion through acquisitions entails the same problems that we encountered in the Caspian Sea project valuation. For example, on March 20, 2003, Cisco announced plans to acquire The Linksys Group Inc. of Irvine, California, for a total purchase price of approximately \$500 million. Although we do not have detailed information concerning Cisco’s analysis of the acquisition, the following discussion (based on a company press release)⁶ highlights how Cisco described its assessment of the five basic issues identified earlier.

Issue #1: Does the “story” make sense?

“Fueled by consumer broadband adoption, the home networking space has experienced mass market acceptance. Linksys has captured a strong position in this growing market by developing an extensive, easy-to-use product line for the home and small office.”⁷ The acquisition is an example of Cisco’s strategy to broaden its end-to-end portfolio of network solutions into high-growth markets such as wireless, voice-over-IP, and storage-area networking.

Issue #2: What are the sources of risk in the investment, and can these risks be managed?

An important source of risk in the investment relates to technological risks. For example, are there wireless technologies that might develop that would render Linksys’s products noncompetitive? This risk is very real for technology-based companies; however, Cisco has partially mitigated this risk by acquiring the recognized industry leader. The challenge

⁶ Cisco Systems press release, March 20, 2003, http://newsroom.cisco.com/dlls/corp_032003.html.

⁷ “Cisco Systems Announces Agreement to Acquire the Linksys Group, Inc.,” news release, March 20, 2003.

facing Cisco in the future will be to maintain Linksys's competitive edge while operating the company as a division of a much larger firm.

Issue #3: How can the investment be financed?

"Under the terms of the agreement, Cisco will issue common stock with an aggregate value of approximately \$500 million to acquire the Linksys business and to assume all outstanding employee stock options."

Issue #4: What is the short-term effect of the investment's acceptance on the firm's reported earnings?

"Exclusive of acquisition charges, Cisco anticipates this transaction will add approximately \$0.01 to its FY2004 pro forma EPS. The transaction will be accretive to both GAAP and pro-forma earnings thereafter."

Issue #5: Does the investment have inherent flexibilities that allow the firm to modify it in response to changing circumstances? In particular,

Can the investment be staged?

- Cisco's acquisition of Linksys is another step in a broad strategy to expand its home and small office product line aimed at developing a dominant position in the home and small office networking market.

Does the investment offer the opportunity for follow-on investments?

- "This acquisition represents Cisco's entry into the high-growth consumer/SOHO [small office, home office] networking market. Home networks allow consumers to share broadband Internet connections, files, printers, digital music, photos, and gaming, all over a wired or wireless LAN (local area network)."

Does the acquisition offer production and/or marketing synergies with existing products?

- Clearly, Linksys offers products that can be marketed using channels of distribution similar to those already in place at Cisco. In addition, similarities between small business and home applications should provide synergies with Cisco's existing marketing assets.

4 DEALING WITH COMPLEXITY—PROCESS AND DISCIPLINE

Whether an investment is like the Caspian Sea investment or like Cisco's acquisition of The Linksys Group, its evaluation can become very complex. To address this complexity in a disciplined way, firms develop policies and procedures that prescribe how to evaluate new investment opportunities. The purpose of these procedures is to ensure that projects receive a thorough analysis and that the project selection process is not subverted by the special interests of one or more managers.

The Investment Evaluation Process

Figure 4 summarizes an **investment evaluation process**. This three-phase process captures the critical elements of project *or* enterprise evaluation, beginning with idea

Figure 4**Three-Phase Investment Evaluation Process—Covering All the Bases**

PHASE I	Investment (Idea) Origination and Analysis
Step 1:	Conduct a strategic assessment.
Step 2:	Estimate the investment's value ("crunch the numbers").
Step 3:	Prepare an investment evaluation report and recommendation to management.
PHASE II	Managerial Review and Recommendation
Step 4:	Evaluate the investment's strategic assumptions.
Step 5:	Review and evaluate the methods and assumptions used to estimate the investment's NPV.
Step 6:	Adjust for inherent estimation errors induced by bias, and formulate a recommendation regarding the proposed investment.
PHASE III	Managerial Decision and Approval
Step 7:	Make a decision.
Step 8:	Seek final managerial and possibly board approval.

generation and ending with a final go–no go investment decision. The process is very general and illustrates how investments are evaluated across a wide variety of industry settings and sizes of firms. Be aware that in this discussion, the phases can be broad enough to cover enterprise evaluation or narrow enough to cover project valuation.

Phase I: Investment (Idea) Origination and Analysis

Firms learn about major investment opportunities from a variety of sources. A common source of ideas is the firm's employees and customers. Another source is outside organizations that bring proposals to the firm's attention, such as investment bankers and other firms that want to buy or sell specific assets.

This initial phase of the analysis includes three activities:

Step 1: Conduct a strategic assessment. Every investment opportunity has an underlying strategy ("value proposition") that provides the basis for making the investment. The initial screening of investment proposals begins with an assessment of the soundness of this underlying strategy. Members of the firm's business development group often do this screening. If the investment (project or enterprise) looks promising after the initial review, then an investment evaluation team further investigates the opportunity and makes a recommendation to management.

Step 2: Estimate the investment value ("crunch the numbers"). The objective of the estimation part of the process is to determine whether the investment has the potential to create value for the firm's stockholders. Traditionally, this analysis involves applying valuation models such as discounted cash flow and market-based multiples (e.g., market to book, price to earnings). However, when the investment's

potential to create value has multiple sources, the evaluation can involve many other types of analysis. These include analyses of (a) projected cash flows from existing or proposed operations; (b) built-in project flexibility that enables management to modify the investment over its life in response to changing market conditions (i.e., real options); (c) opportunities to manage (i.e., control or even eliminate) some of the investment's inherent sources of risk; (d) the ability to structure the organization (align decision authority and compensation policies) to enhance the investment's value-creation potential; (e) financing alternatives; and (f) the effect of the investment on the firm's near-term earnings.

Step 3: Prepare the investment evaluation report and recommendation to management.

Combining the strategic and quantitative analyses from Steps 1 and 2, the investment analysis team prepares a report summarizing its recommendation to the managerial oversight committee. At a minimum, the report will contain (a) an assessment of the investment's strategy for creating value and the firm's comparative advantage in carrying out the strategy, (b) an estimate of the value of the investment (net present value), and (c) the supporting information and assumptions used in the analysis.

Phase II: Managerial Review and Recommendation

Managerial review serves a control function: It screens the initial analysis for potential sources of errors and investment-selection bias. A different group of employees, ones who did not perform the original analysis, review the new investment proposals. Typically, firms use an investment review committee (sometimes known as the *strategic planning committee*) to review new proposals.

The fundamental responsibility of the investment review committee, as the name implies, is to review the recommendation of the initial investment analysis team. The review committee will make sure that the assumptions of the initial evaluation team are reasonable and that nothing has been forgotten in the analysis. If the review process recommends the investment to management, the review committee also assesses how much to invest. The investment review committee provides a system of checks and balances designed to weed out bad investments and flawed investment analysis.

Phase II involves the activities of Steps 4 through 6:

Step 4: Evaluate the investment's strategic assumptions. The review process, like the initial investment screening (Step 1), begins with an assessment of the investment's value proposition or strategy. Does the story underlying its value-creation potential make sense to the managerial review committee? Do the hedging opportunities offer sources of value to the firm? Do investment-specific financing alternatives make sense, and do they have a reasonable prospect of adding value?

Step 5: Review and evaluate the methods and assumptions used to estimate the NPV of the investment. The committee evaluates the quantitative analysis conducted in Step 2. Are the assumptions that underlie earlier price, cost, and quantity estimates reasonable in light of observed market prices and the market shares of competitor firms and products? Are there any additional options (follow-on investments and other important sources of value creation) inherent in the proposal that were not considered in the earlier report? Finally, are there opportunities to expand or contract the scope of the investment (even abandon it altogether)? Although it can be very difficult to estimate their value, it is critical that the firm consider these options in order to properly value the investment.

Step 6: Adjust for inherent estimation errors induced by bias, and formulate a recommendation regarding the investment. The origination process can be fraught with the potential for biased estimates of an investment's value. After making appropriate adjustments, the managerial review committee passes its recommendation along to the executive responsible for making a final decision.

Phase III: Managerial Decision and Approval

The final responsibility for making the investment rests with an executive in the firm who possesses the appropriate level of authority. Typically, the larger the financial commitment an investment requires, the higher the level in the management hierarchy that is required for commitment approval. Consequently, in the final phase of investment evaluation, the recommendations of the review and evaluation team go to the firm's management team. The management team approves or rejects the proposed investment, and seeks board approval if necessary.

Phase III entails the final two steps in our eight-step investment evaluation process:

Step 7: Make a decision. Combining top management's sense of the firm's overall business strategy with the recommendation of the investment review committee, the executive with the appropriate level of authority makes a decision. The choices are to reject the proposal outright or to accept the proposed investment for immediate or deferred implementation.

Step 8: Seek final managerial and possibly board approval. If the decision is to undertake the investment, then the decision maker seeks final approval from the firm's management team and possibly from the board of directors.

The three-phase investment evaluation process provides a complete and integrated analysis of all facets of investment valuation, while maintaining the critical control features. However, as firms implement this process in real-world cases, a range of issues can arise to bias the results. To demonstrate the use of this evaluation process and show some of the complexities that can arise, we conclude the chapter with a brief case study.

5 CASE STUDY—CP3 PHARMACEUTICALS LABORATORIES INC.

Figure 4 presented a three-phase, eight-step process for carrying out a disciplined approach to the investment evaluation and decision process. The need for such an approach was evident in the very large and complex Caspian Sea investment project. Firms can—and should—use the same basic approach for making smaller and more focused investments, as well as for acquiring entire enterprises.

In this section, we will use a hypothetical example to walk through a typical investment evaluation for a relatively small project. It involves the desire of CP3 Pharmaceuticals Laboratories Inc. (a fictional name for a real company) to invest \$547,000 to install a new materials-handling system. The example begins with the identification of the idea and concludes with the final go–no go decision.

As you read the description of the decision-making process, keep in mind the three-phase, eight-step investment evaluation process outlined in Figure 4. CP3's investment decision-making process does not line up exactly with the process illustrated there. Such

differences are not unusual, because firms often modify their processes to fit their own particular needs. However, the process does touch on each step.

Example: Investing in a New Materials-Handling System

Business practice often deviates from theory for reasons that are understandable. As you read through the following fictional account of an investment by CP3 Pharmaceuticals Laboratories Inc., focus on the process and not the numbers. Note specifically how the investment is being analyzed. If you have studied finance in the past, you may even observe some instances where the theory you have been taught conflicts with practice.

Susan Chambliss is a vice president for business development at CP3 Pharmaceuticals Laboratories Inc., located in Austin, Texas. The members of CP3's business development group scout for new investment opportunities for the company on an ongoing basis. Most firms of any size have such a group. Investment opportunities for CP3 could include new products or new markets for existing products. Therefore, it is not at all unusual that someone from this group, Susan in this instance, identifies a promising investment opportunity.⁸

CP3 operates a medical-packaging operation at its Austin facility that has significant costs from product waste. Susan has identified a new materials-handling system that offers the potential for substantial cost savings through waste reduction, reduced head count in the manufacturing area, and savings through plastic recycling. Figure 5 provides details of the proposed project, in the format of the company's capital expenditure request form.

Before initiating a formal analysis of the proposed investment, however, Susan engages in informal discussions with senior executives on CP3's strategic planning committee. This committee reviews all major investment proposals that the firm undertakes and makes a recommendation to the firm's management as to whether or not a project is viable. The makeup of this committee at CP3 is typical of such committees at other companies: It includes the company treasurer, the chief financial officer (CFO), the chief operating officer (COO), and the chief executive officer (CEO).

Because Susan will ultimately need the approval of this committee, it makes sense that she begin by floating her idea, while still in the formative stage, by one or more of the members. Given the time and effort associated with the analysis of a new investment opportunity, she will want a preliminary indication that the project has a reasonable chance of approval before going forward. For example, it might be that company executives are considering the possible closure or sale of the medical-packaging operations. If so, then it clearly would not make sense for Susan to invest time or energy to form a working group to explore her idea further.

After several informal discussions, Susan concludes that the new materials-handling system has promise, so she initiates the study. For a project as small as this one, Susan simply assigns the project to a single financial analyst. For very large and complex projects, she might form a team consisting of several people who possess the requisite skills to understand and evaluate the investment opportunity.

⁸ This does not mean that all investment ideas originate out of business development. To the contrary, in a healthy firm, investment ideas come from all over the firm. However, at some point, if the investment is significant, the business development group will get involved.

Figure 5 Capital Expenditure Request Form**1. Executive Summary**

CP3 Pharmaceuticals Laboratories' Austin, Texas, plant is requesting \$547,000 to purchase and install a new scrap materials-handling system for its medical-packaging operations.

Purchasing the new system will allow the plant to meet the following objectives:

- Reduce waste in the firm's packaging operations, for savings of \$300,000 per year.
- Reduce head count from the test area. Estimated savings of \$35,000 per year.
- Recycle plastic materials that historically were part of waste, with disposal cost of \$8,800 per year.
- Earn a 20% rate of return on invested capital.

**2. Proposal and Justification**

CP3's medical-packaging unit is expected to produce over 400 million vials of over-the-counter drugs this year. The packaging of these vials will generate 1.5 million pounds of scrap plastics. Of this total, one-third can be recycled, and the remainder becomes scrap. Under the present method, the scrap is collected in bins at the end of each of six production lines. The bins are then collected every 15 minutes and transferred to a grinding room, where the scrap is either ground for resale or transported to trash. At present, the disposal cost for the 1 million pounds of scrap plastic is \$8,800 per year.

The proposed scrap materials-handling system involves placing a small grinder at the end of each production line that can grind the plastic, then sending the scrap via vacuum tubes to a scrap collection site in another part of the plant. The ground-up scrap can then all be sold for \$300,000 per year, while eliminating the scrap-disposal cost of \$8,800 per year.

3. Financial Analysis

	2005	2006	2007	2008	2009	2010
Capital spending	(547,000)					
Impact on revenue/(expense)						
Scrap revenue		300,000	300,000	300,000	300,000	300,000
Labor savings		35,000	35,000	35,000	35,000	35,000
Reduced recycle costs		8,800	8,800	8,800	8,800	8,800
Total impact		343,800	343,800	343,800	343,800	343,800
Less: Depreciation (5 years)		(109,400)	(109,400)	(109,400)	(109,400)	(109,400)
Net operating income b/tax		234,400	234,400	234,400	234,400	234,400
Less: Taxes (40%)		(93,760)	(93,760)	(93,760)	(93,760)	(93,760)
Net operating profit after tax		140,640	140,640	140,640	140,640	140,640
Plus: Depreciation expense		109,400	109,400	109,400	109,400	109,400
Less: Capital expenditures	(547,000)	—	—	—	—	—
Project free cash flow (Project FCF)	(547,000)	250,040	250,040	250,040	250,040	250,040
Net present value (NPV)	\$200,773					
Payback (in years)	2.19 years					
Internal rate of return (IRR)	35.8%					

4. Risks

The grinders and vacuum transport systems have been tested for over a month with only two minor failures. The failures relate to stoppages at the end of line 4, which carries some of the larger scrap pieces. This problem has been addressed by increasing the size of the grinder on that line and by installing sensors to provide an alert that a stoppage is about to occur so the operator can stop the process and clear the grinder and vacuum tubes.

5. Project Timeline

It will take three months to get the new system up and running, because the installation must work around production shifts already in place.

Susan asks the analyst to prepare a report for formal submission of the project to the strategic planning committee. The information in Figure 5 is an abbreviated version of a firm's typical investment evaluation report. The report provides a thorough analysis of the project. It begins with a list of the various reasons why the group believes the project is likely to be successful. In addition, it includes summary measures of project value, reported in the measures the company uses—in this case, net present value, internal rate of return, and payback. (These are all concepts that we review in later chapters.) The financial analyst who prepared the analysis also included specific cost and cash flow projections to back up the summary analysis. These estimates span a period of five years, ending in 2010.

If this project were larger, the report would probably address several other important issues as well. For example, it might include any or all of the following: an estimate of the expected impact of the investment on CP3's reported earnings for several future quarters; an analysis of various scenarios involving key cost and revenue drivers, to highlight the risks inherent in the investment; and a discussion of how the firm can finance the project, as well as risk-management issues.

Once the project analysis report goes to a strategic planning committee, this group will often have another analyst prepare an independent assessment of the proposal's merits. This review checks the assumptions and methodology of the original project valuation estimate. A key concern with regard to the integrity of the investment evaluation process is that the project review and analysis group be truly independent of the analysts who prepared the initial project report and of the analysts assigned by management to review it. Furthermore, the strategic planning committee must be given the authority and funding needed to perform an independent appraisal of the project.

Addressing the Possibility of Decision Bias

Biases of various sorts can enter into the analysis of new investment proposals. Using the CP3 case as an example, Susan and members of the project analysis team may be biased if they have *incentives* to get the deal approved. Indeed, it is Susan's job to identify good opportunities for the firm, and her bonus and her ability to keep her job may depend on her ability to get her ideas approved. Bias may also enter the process simply because of *human nature*: Psychologists have found that individuals tend to be overconfident and overly optimistic about their own ideas.

The Role of the Strategic Planning Committee As the Skeptical Boss

In light of the potential for bias in favor of new investments that make their way up to the strategic planning committee for review, it becomes the task of this committee to play the role of a skeptical boss. Very simply, this committee has the task of reviewing major investment proposals and attempting to ferret out any bias in their analysis that arises out of the natural tendency for project champions to be overconfident and overly optimistic.

The members of the strategic planning committee are experienced in evaluating projects. They understand that projects are often completed at 50% over budget and are rarely completed more than 10% under budget. They also understand that realized rates of return are almost always lower than projected rates of return. With this in mind, the strategic planning committee's role requires that they carefully work through the assumptions and the analysis and that they question everything. If the controls in the firm

are functioning properly, the strategic planning committee must sign off on the analysis before the project can go forward.

Although the strategic planning committee will carefully review the analysis on which the project is based, its members may also have a broader perspective. They may consider issues that go beyond the attributes of this specific project. For example, they might be considering the possibility of moving the business offshore and closing down the Austin plant, in which case the new project makes little sense.

Finally, the strategic planning committee is likely to carefully consider the firm's alternatives for financing the project. If the project can easily be funded internally, it is more likely to be approved than if it requires external funding. If the project were so big that it would require the firm to issue equity, the ultimate approval for the project might depend on whether the firm's top management believed this a good time to issue equity—a decision that has nothing to do with the particular attributes of the project.

If the strategic planning committee suggests approval, it then sends the proposal to the executive who has sign-off authority on the capital expenditure required to fund the project. In the case of the scrap materials-handling system, CP3's CEO is the decision maker. The CEO is likely to consider the same issues that the strategic planning group considered. He or she will, of course, have less time to review the specifics of the proposal than did the strategic planning committee and will rely on the analysis done for that group.

If the project is so large that it requires approval by the board of directors, the CEO will then take the project to the board. Generally, if the project has the backing of the firm's executive, the board is unlikely to turn down the project, but board members may question some of the aspects of the contractual structure. For example, directors may question the compensation and governance structure of the deal. They may also question how the project is financed, particularly if the project requires an external-equity issue.

6 SUMMING UP AND LOOKING FORWARD

Final Comments—The Investment Decision-Making Process

Our discussion of the investment decision-making process gives rise to a number of important observations that will influence the content and structure of the rest of this book:

- *The process can be very costly.* The process of project origination, evaluation, and approval is expensive and time-consuming. Of course, skimping on the analysis of major projects can be even more expensive if it leads to project failures or missed opportunities.
- *The process can be subject to biased estimates of project value arising out of conflicts of interest and incentive problems.* Individuals within the process often have conflicting motives. For example, it is natural for members of the team that champions a proposal through the approval process to become personally committed to the success of the project, and, as a result, their analyses may become biased. Furthermore, there is often a financial incentive attached to getting the project approved. For example, year-end bonuses may be tied to getting “deals done.” This incentive can easily lead employees to portray a project's prospects in a more optimistic way than may be warranted. On the other hand, the members of the various units within the firm who must analyze and “sign off” on the project are often skeptics. They typically are staff personnel whose role is to ferret out bias in the investment analysis and provide a control over overzealous project champions. Members of the internal

control group are not rewarded for deal completion but may suffer consequences when they sign off on a project that fails.

- *The process is affected by problems arising out of differences in the information available to project champions and the internal review or control group (the strategic planning committee in our earlier example).* Specifically, the control group in Phase II of the process is generally less well informed about the project's inner workings than the project proponents from Phase I. In the interest of efficiency, this situation would ordinarily support delegation of decision-making authority to the managers and project champions who know the most about the project. However, in reality, incentive issues and the natural bias that project champions often exhibit require that some type of control system (such as the strategic planning and review committee) be put in place, and this is exactly what we see in business practice.

An unbalanced emphasis on any one of the three phases of project evaluation shown in Figure 4 can have disastrous effects on the firm. For example, too much emphasis on the investment origination (idea) phase, to the detriment of managerial review, can lead to the firm adopting questionable project investments that have a low probability of success. Equally dangerous is overemphasis on the managerial review (Phase II) relative to the investment origination phase. In this instance, excessive caution keeps the firm from undertaking promising new investments. Finally, overemphasis on the final phase of the analysis can result if executives try to micromanage the firm. This can stifle the initiative of employees working in the origination and review phases. In an extreme case, the origination and review phases become nothing more than attempts to “second-guess” the preferences of the firm's top manager(s).

Looking Forward

Academics who study finance sometimes strip away some of the complexities of corporate investment decisions in an effort to focus on the heart of what determines value. However, by doing this, we can create a disconnect between what *should* be done in theory and what *is* in fact done in practice. For example, academics have very well-developed theories that describe how firms should determine the **discount rates** that they use to evaluate investment projects. To a large extent, corporate executives are aware of these theories but have corporate **hurdle rates** (i.e., the minimum rate of return on acceptable projects) that often greatly exceed the discount rates suggested by academic theories. These high hurdle rates may be used to counteract the overoptimism of the managers, or they may provide ambitious targets that serve to motivate the managers. Throughout this book, we will emphasize *both* the fundamentals of valuation, as described by academic theories, and the real-world complexities that make industry practice somewhat different.

The last three decades have provided a number of advances in the understanding of the economic forces that drive investment valuation. While we will be discussing recent advances in a variety of aspects of investment valuation, we will pay particular attention to how recent innovations in the valuation of derivative securities, like options, are used to value real investment projects and investment strategies.

7 SUMMARY

Valuation is more than discounting cash flows. The evaluation of new investment opportunities, ranging from the smallest capital budgeting exercise up to the acquisition of an entire firm, has come a long way since the early days in which the focus was solely on the present value of estimated future cash flows. Although the importance of net present value (NPV) has not changed, analysts now consider a broad set of factors. For example, when a firm analyzes whether to make a significant new investment, it will generally consider the following issues:

- *Cash flow estimation.* The first step in valuing an investment is to estimate future cash flows. One must ask how certain these cash flow estimates are, as well as how sensitive the estimates are to unexpected changes in the economic environment.
- *Risk assessment.* Major new investments by multinational firms often involve committing funds to emerging markets. Such investments typically come with significant additional risks: political risk, commodity price risk, interest rate risk, and exchange rate risk. Risk must be assessed as well as managed, and it plays an important role in determining the rate at which cash flows are discounted as well as the financing of the investment project.
- *Financing opportunities.* Financing can offer an important source of value for an investment project and can be a key determinant of the investment's cost of capital.
- *The effects on the firm's near-term earnings.* Although the value of an investment is created by the cash flows generated by the investment, managers are keenly aware of how a project affects earnings. In practice, an investment's effect on the earnings of a firm will have a large influence on whether the investment is ultimately taken.
- *Staged investments.* The decision to initiate an investment essentially involves acquiring the "option" to invest in later stages of the project (assuming the intermediate stages prove fruitful). Thus, analysts can use option valuation techniques to evaluate whether to postpone, speed up, or cancel future investments.
- *"Follow-on" investment opportunities.* Previous investments are a primary source of new investment opportunities. Thus, it is important that we consider the impact of major investment projects on future opportunities.

We recognize at the outset that the analysis of investments is difficult to get right. Even well-intentioned managers can make mistakes, either because of psychological biases that tend to make them overly optimistic or because project valuations pose difficult problems that are not always well understood. Consequently, it is essential that the firm have in place a well-designed process that incorporates appropriate decision-making checks and balances. Moreover, we emphasize the need to continually push for quantification, even in the face of difficult issues that make measurement challenging.